

Question

Thermogravimetric analysis was performed on a certain ternary compound **G** containing Be, and 1.000 g of the compound was heated under an inert atmosphere:

- The first weight loss occurred at 80°C, with 0.789 g of solid remaining;
- The second weight loss occurred at 150°C, with 0.683 g of solid remaining;
- The third weight loss occurred at 400°C, with 0.149 g of solid remaining;
- Further heating to higher temperatures resulted in continued weight loss, but it was not significant.

It is known that only two small molecules **A** and **B** are considered to be produced during the weight loss process (**A** and **B** may continue to decompose later, but this is not relevant to the question). Both **A** and **B** are composed of two elements, **X** and **Y**. The ratio of the number of **X** to **Y** in **A** is the same as the ratio of the number of **Y** to **X** in **B**, and the mass fraction of element **X** in **A** does not exceed 3%.

Deduce the element symbols or chemical formulas of **X**, **Y**, **A**, **B**, and **G**, and select the correct option from the following options.

- A.** All other options are incorrect
- B.** The mass fraction of **X** in **B** does not exceed 7 times the mass fraction of **X** in **A**
- C.** Both **A** and **B** are highly soluble in water, and their aqueous solutions exhibit alkalinity.
- D.** The first two instances of weightlessness produced only the same small molecule, whereas the third instance of weight loss yielded two different small molecules.
- E.** The ratio of the numbers of **X** and **Y** in **G** is 4 : 5
- F.** The ratio of the numbers of **X** and **Y** in **G** is 5 : 4

- G.** If **G** is an ionic compound, the ratio of anions to cations is 2 : 1
- H.** If **G** is an ionic compound, then the cation has two different ligands.
- I.** After the third weightlessness, the residual compound is a hydrolyzable binary compound.

Answer

Correct Answer: **G**

Detailed Explanation

The mass fraction of **X** in **A** does not exceed 3%. Assuming the chemical formula of **A** is XY_a (where a may not be an integer), the following relationship can be calculated: (where M represents atomic/molecular weight)

$$\frac{M(\text{Y})}{M(\text{X})} > 32.33 \times \frac{1}{a}$$

This indicates that either $M(\text{Y})$ is much larger than $M(\text{X})$, or a is particularly large. Since **A** is a small molecule, and considering the first three periods of elements, it can be inferred that **X** is most likely **H**, otherwise the formula weight of **Y** would be particularly large, which is obviously not a common small molecule. Therefore, first consider that **X** is **H**.

Next, infer **Y**. Since the first three periods of elements are preferred, it can be known that $a > 1$. Since common small molecules are preferred, and considering that this molecule is volatilized during thermogravimetric analysis, exclude **Y** being a metal. Among the hydrides of nonmetals in the first three periods, the common hydrides that satisfy $a > 1$ and also have both XY_a and X_aY are only HN_3 and H_3N , and HO_2 and H_2O . It is also found that the mass fraction of **H** in HO_2 is $3.05\% > 3\%$, excluding the latter group. Finally, it is deduced that **X** is **H** and **Y** is **N**.

CHECKPOINT

1 PTS

X is **H**, **Y** is **N**

CHECKPOINT

2 PTS

A is HN_3 , **B** is NH_3

Next, analyze options B and C. Calculate that the mass fraction of H in NH_3 is approximately 7.6 times the mass fraction of H in HN_3 , which is greater than 7 times, so B is incorrect. Both HN_3 and NH_3 are easily soluble in water, but the former is weakly acidic when dissolved in water, and the latter is alkaline, so C is incorrect.

Next, perform thermogravimetric analysis. The total weight loss is 0.851 g. Let the amount of substance of HN_3 be m mol, the amount of substance of NH_3 be n mol, and the initial amount of substance of the ternary compound be b mol. Then the following molecular weight relationship can be obtained:

$$\frac{0.851}{b} = m \times 43.03 + n \times 17.01$$

The mass of the three weight losses are 0.211 g, 0.106 g, and 0.534 g, respectively. It can be found that the ratio of the first and second weight loss mass is approximately 2 : 1, which suggests that the two weight losses are the same substance, and the molar ratio is 2 : 1. Try whether this substance is HN_3 or NH_3 respectively, and b can be calculated, and then substituted into the above formula. After enumeration, it can be found that when the substance is NH_3 , $n = 3, m = 2$ is a reasonable solution. Therefore, it is preliminarily determined that this is the inference result.

CHECKPOINT

2 PTS

The first two weight losses are both the loss of NH_3

Specifically, $b = \frac{0.106}{17.01} = 6.223 \times 10^{-3}$ mol, so the molecular weight of **G** is approximately $1.000/b = 160.7$ g/mol

CHECKPOINT

1 PTS

The molecular weight of **G** is approximately 160.7 g/mol

After calculation, it can be concluded that the third weight loss only loses two molecules of HN_3 , so option D is incorrect.

CHECKPOINT

1 PTS

The third weight loss only loses two molecules of HN_3

Subtracting three molecules of NH_3 and two molecules of HN_3 from the total molecular weight, the remaining molecular weight is 23.52 g/mol, which looks like a Be and N, but the valence is not correct, so it can be deduced that there is a slight error in the data, and there should be another H. Therefore, the molecular formula of **G** is $\text{BeH}_{12}\text{N}_{10}$, or $\text{Be}(\text{NH}_3)_4(\text{N}_3)_2$.

CHECKPOINT

2 PTS

The molecular formula of **G** is $\text{Be}(\text{NH}_3)_4(\text{N}_3)_2$

From this, it can be known that the ratio of H to N is 6 : 5, so options E and F are both incorrect. It is also known that the product after the third weight loss is $\text{Be}(\text{NH})$, so option I is incorrect.

From the composition of the substance, it is known that if **G** is an ionic compound, the cation should be $[\text{Be}(\text{NH}_3)_4]^{2+}$, with only one ligand. The anion is N_3^- . The ratio of cation to anion is 2 : 1. Therefore, option G is correct and option H is incorrect.

CHECKPOINT

1 PTS

If **G** is an ionic compound, the cation is $[\text{Be}(\text{NH}_3)_4]^{2+}$, the anion is N_3^-

In summary, option G should be selected.